

ABAQUS Fortran Compiler Integration Guide

Intel OneAPI and Visual Studio Configuration Manual
Central Computer & Informatics Centre (CIC)
Indian Institute of Technology Kharagpur

This document provides detailed instructions for integrating **ABAQUS** with Intel OneAPI and Microsoft Visual Studio for compiling Fortran user subroutines on Windows systems.

Important Note: Install all software components in the recommended order. Administrator privileges may be required during installation and configuration.

Compiler Installation

Step 1: Install Microsoft Visual Studio 2017 or 2019.

Please refer to the table below to determine the supported combination for the version of Abaqus you are using:

Qualified Intel Compiler Requirements Matrix for Abaqus / 3DEXPERIENCE

Windows				
Abaqus Version	Intel Packaging Name / Version	Fortran Classic (ifort) Version	Microsoft Visual Studio Version	C++ Compiler/Linker Version
2025	OneAPI HPC Toolkit 2022.1	2021.6.0	2022	14.34.31944.0
2024	OneAPI HPC Toolkit 2022.1	2021.6.0	2022	14.34.31944.0
2023	Parallel Studio XE 2019 Update 5	19.0.5.281	2019	14.28.29913.0
2022	Parallel Studio XE 2019 Update 5	19.0.5.281	2019	14.28.29913.0
2021	Parallel Studio XE 2019 Update 5	19.0.5.281	2017	14.16.27031.1
2020	Parallel Studio XE Composer Edition 2016 Update1	16.0.1	2017	14.16.27031.1

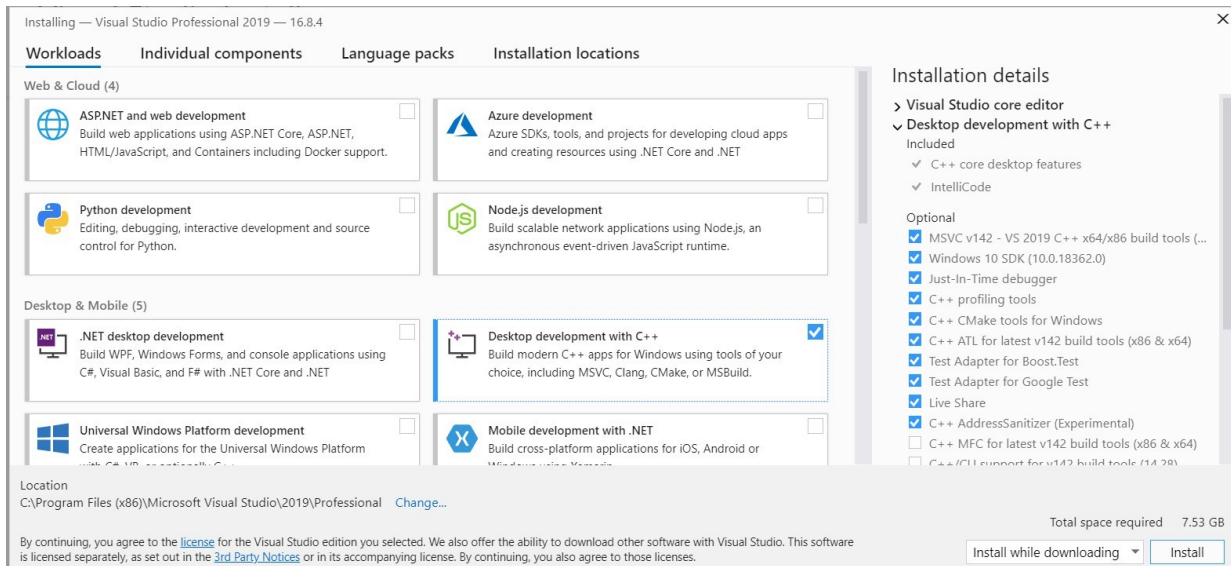
Step 2: Install Intel oneAPI Base Toolkit.

Please refer to the table below to determine the supported combination for the version of Abaqus you are using:

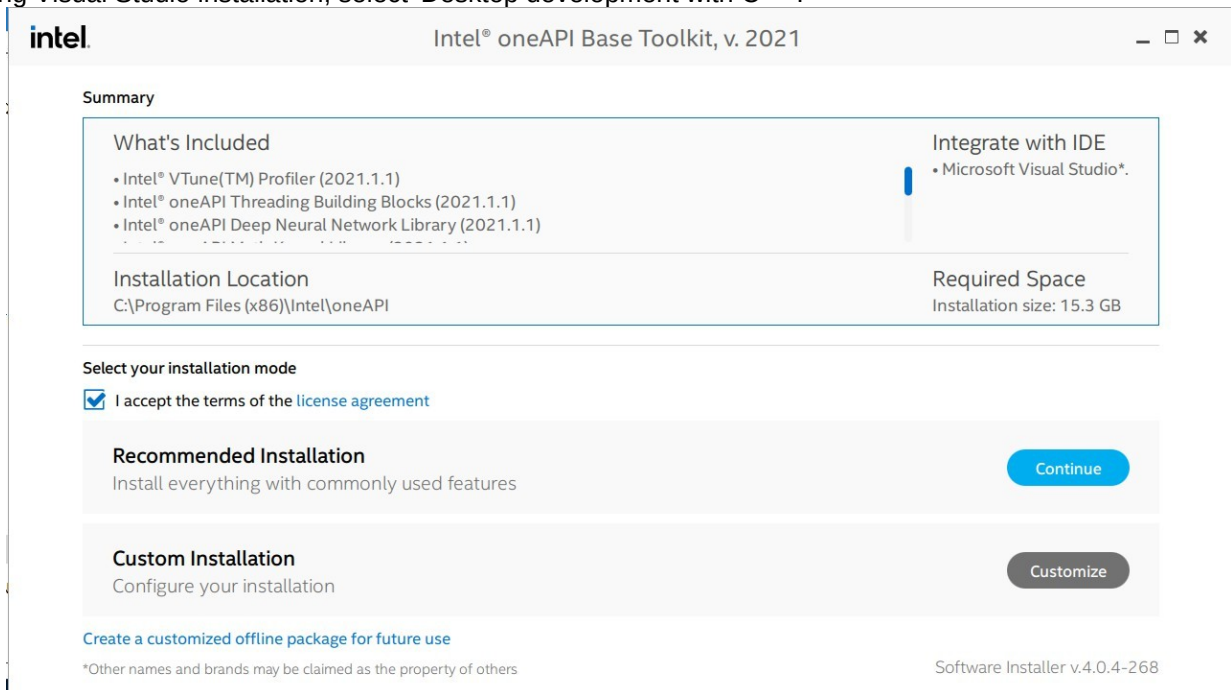
Qualified Intel Compiler Requirements Matrix for Abaqus / 3DEXPERIENCE

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2020	Parallel Studio XE Composer Edition 2016 Update1	16.0.1	2017	14.16.27031.1

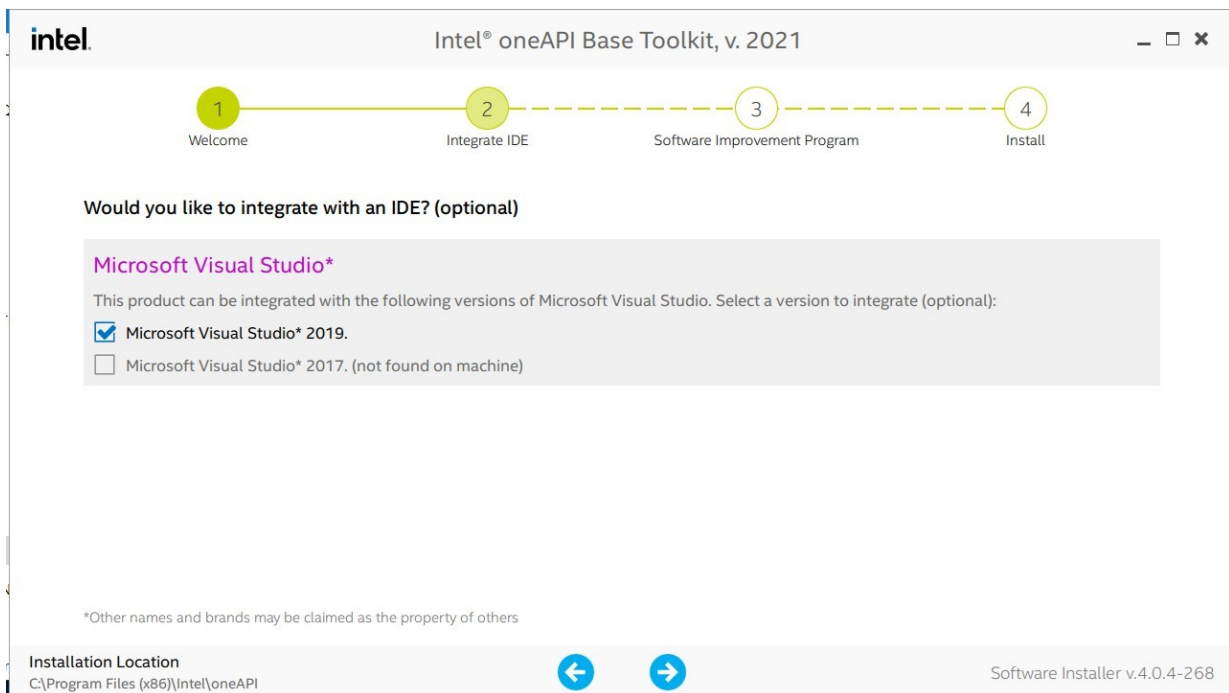
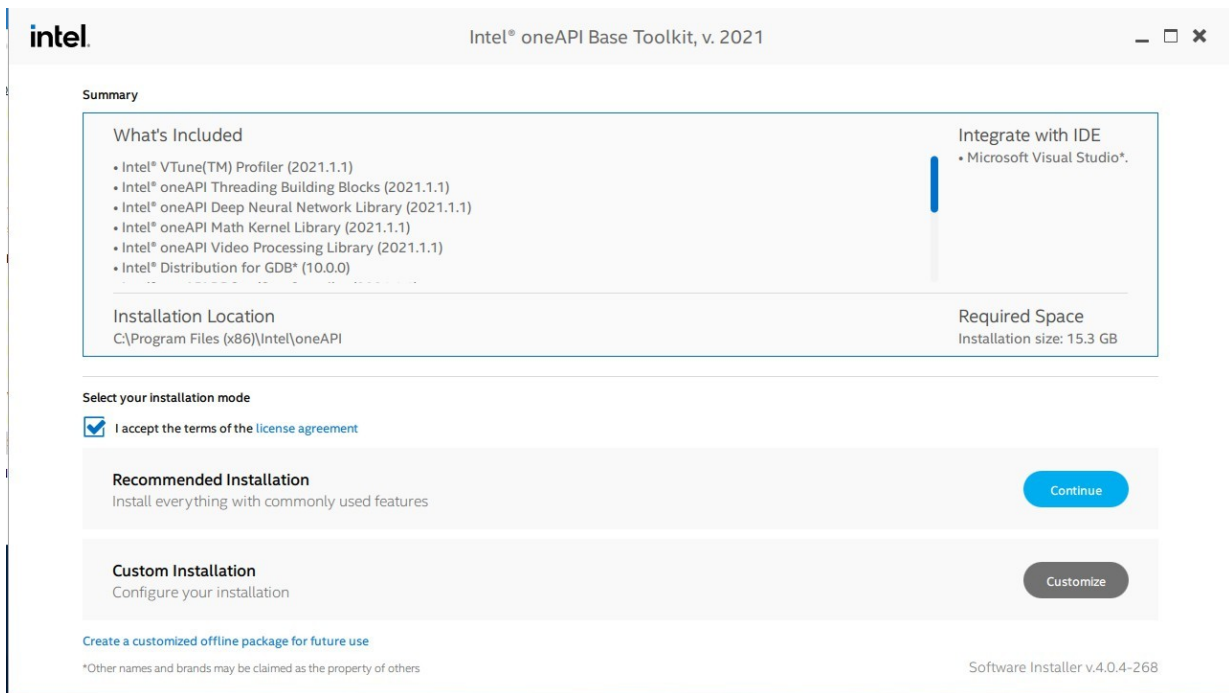
Step 3: Install Intel oneAPI HPC Toolkit.

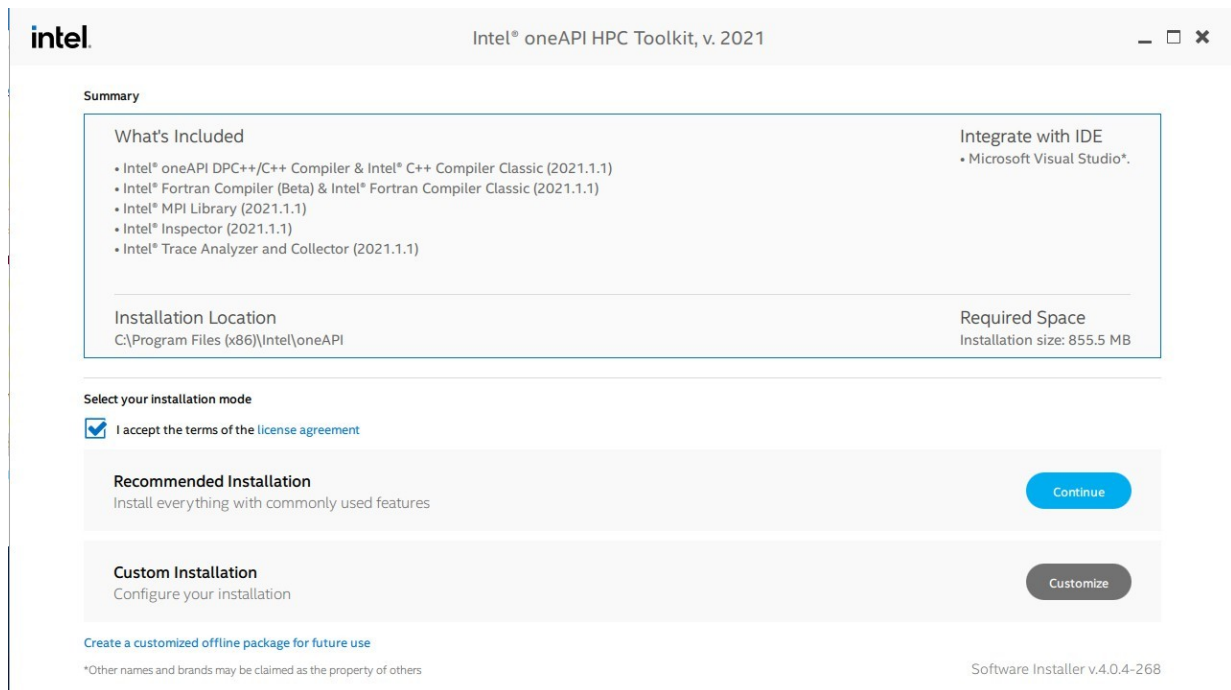


During Visual Studio installation, select 'Desktop development with C++'.



Intel OneAPI installation detects Visual Studio integration automatically.

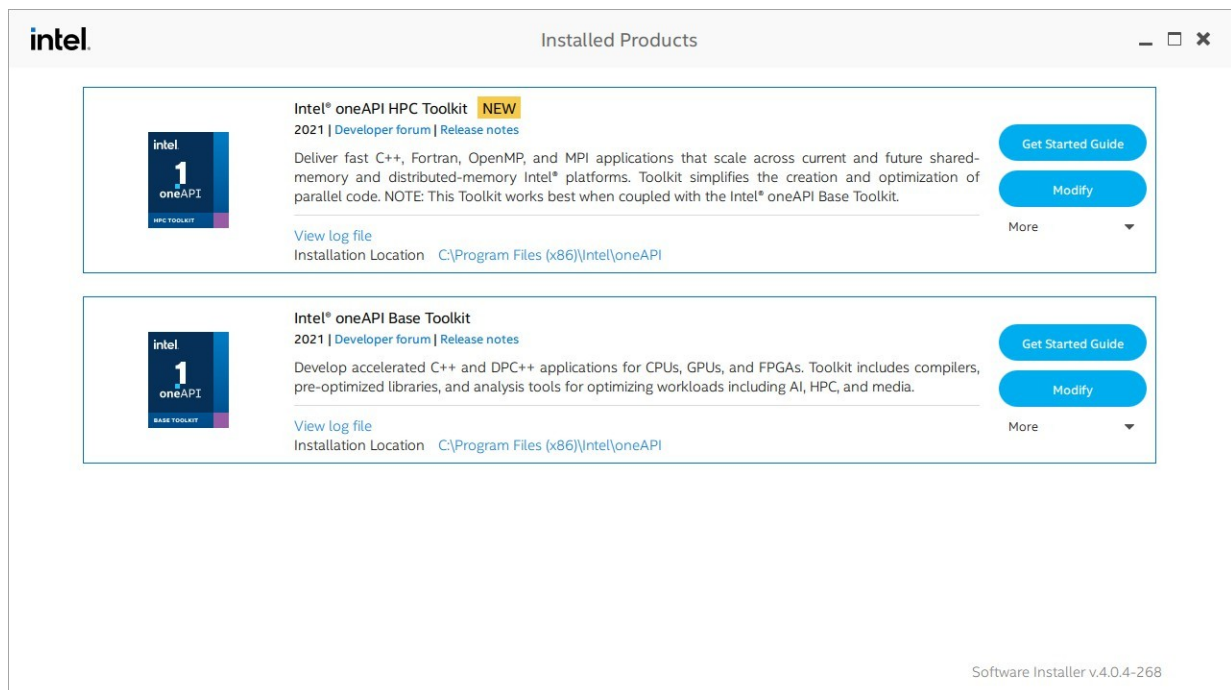




Development Environment Setup

Open Intel oneAPI Command Prompt or execute:

C:\Program Files (x86)\Intel\oneAPI\setvars.bat



The oneAPI environment variables are active only within the current terminal session.

ABAQUS Environment File Modification

Create an abaqus_v6.env file and add:
compile_fortran += ['/names:lowercase',]

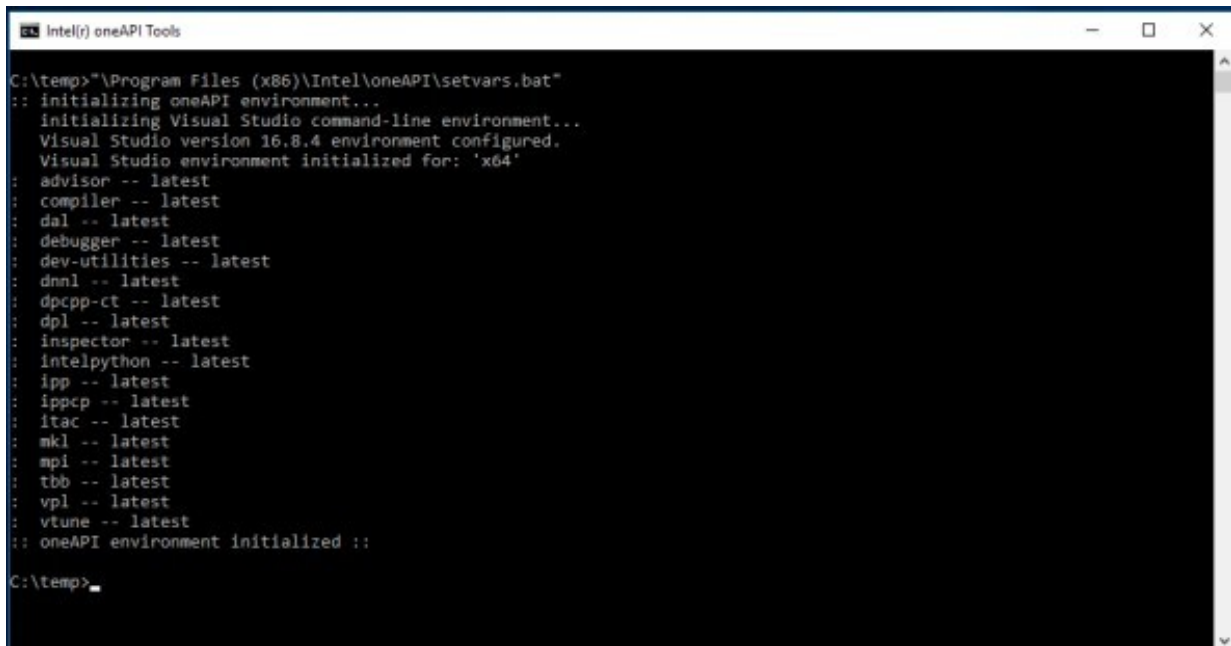
If solver abort occurs, also add:

```
link_sl='LINK /NODEFAULTLIB:LIBCMT.LIB /dll /def:%E /out:%U %F %A %L %B'
```

ABAQUS User Subroutine Verification

Run:

```
abaqus verify -user_std
```

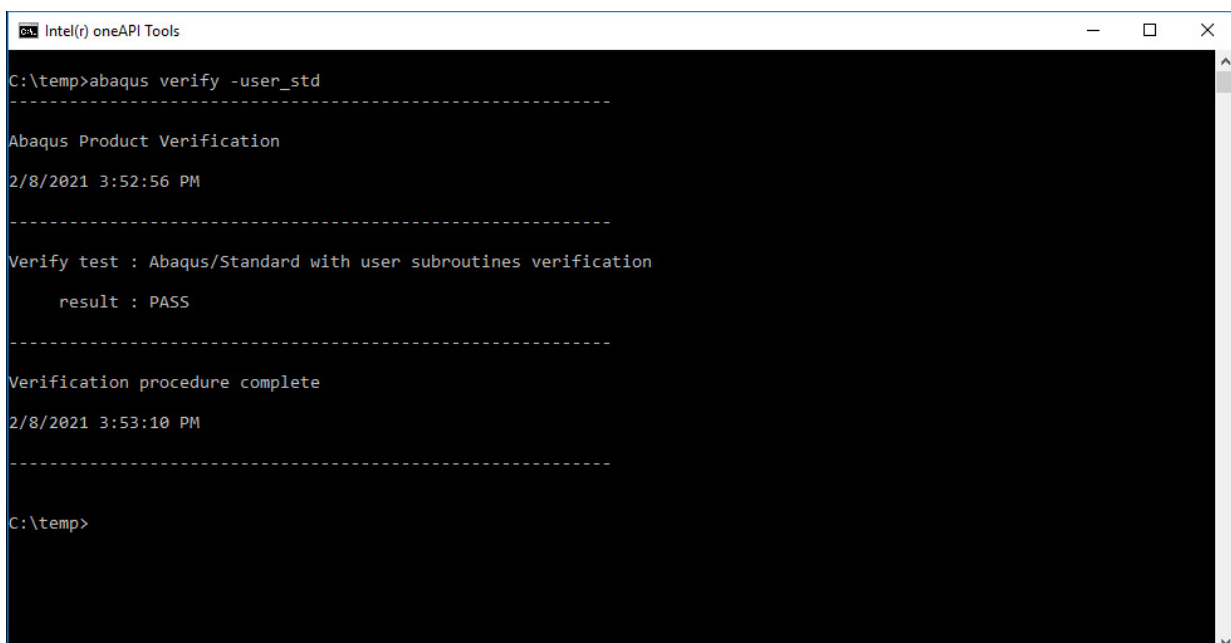


```
Intel(r) oneAPI Tools
C:\temp>"\Program Files (x86)\Intel\oneAPI\setvars.bat"
:: initializing oneAPI environment...
initializing Visual Studio command-line environment...
Visual Studio version 16.8.4 environment configured.
Visual Studio environment initialized for: 'x64'
: advisor -- latest
: compiler -- latest
: dal -- latest
: debugger -- latest
: dev-utilities -- latest
: dnnl -- latest
: dpcpp-ct -- latest
: dpl -- latest
: inspector -- latest
: intelpython -- latest
: ipp -- latest
: ippcp -- latest
: itac -- latest
: mkl -- latest
: mpi -- latest
: tbb -- latest
: vpl -- latest
: vtune -- latest
:: oneAPI environment initialized ::

C:\temp>
```

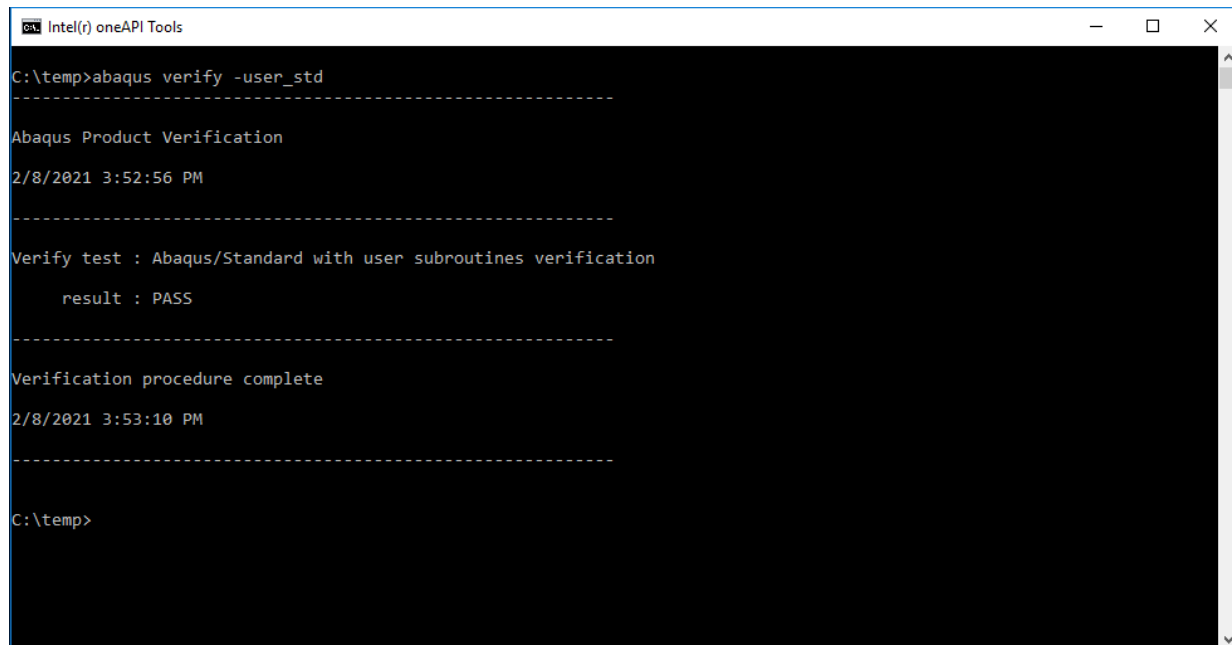
Configuring ABAQUS Start Menu Shortcuts

Step 1: Open the file location of the ABAQUS Command shortcut.



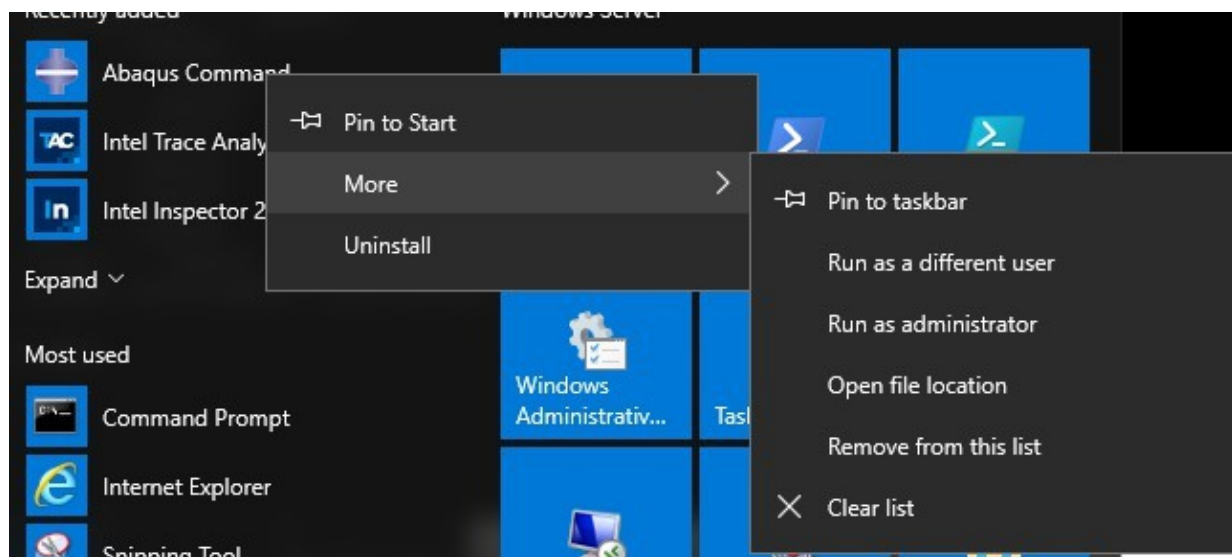
```
Intel(r) oneAPI Tools
C:\temp>abaqus verify -user_std
-----
Abaqus Product Verification
2/8/2021 3:52:56 PM
-----
Verify test : Abaqus/Standard with user subroutines verification
      result : PASS
-----
Verification procedure complete
2/8/2021 3:53:10 PM
-----
C:\temp>
```

Step 2: Open shortcut properties.

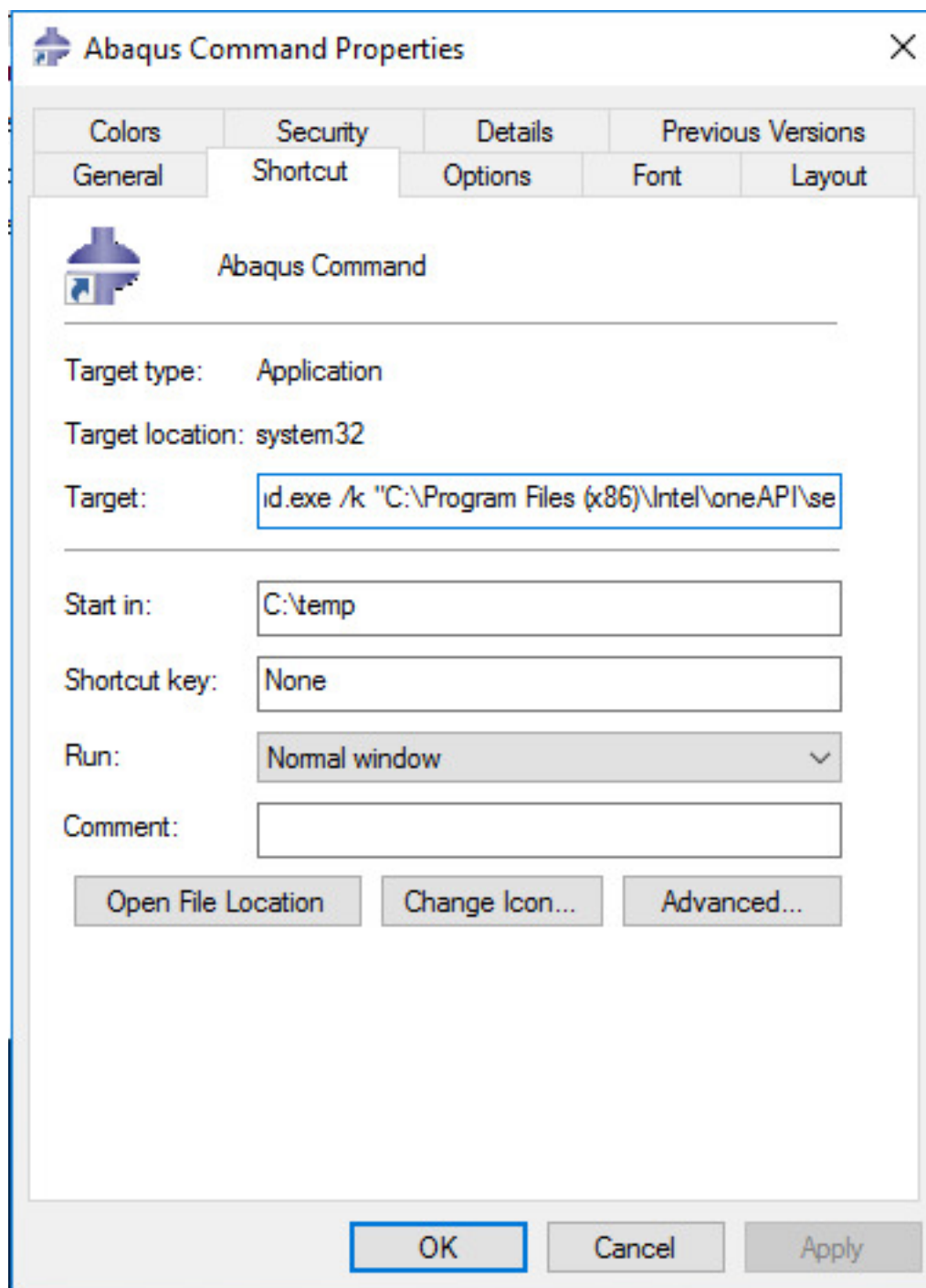


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C:\temp>
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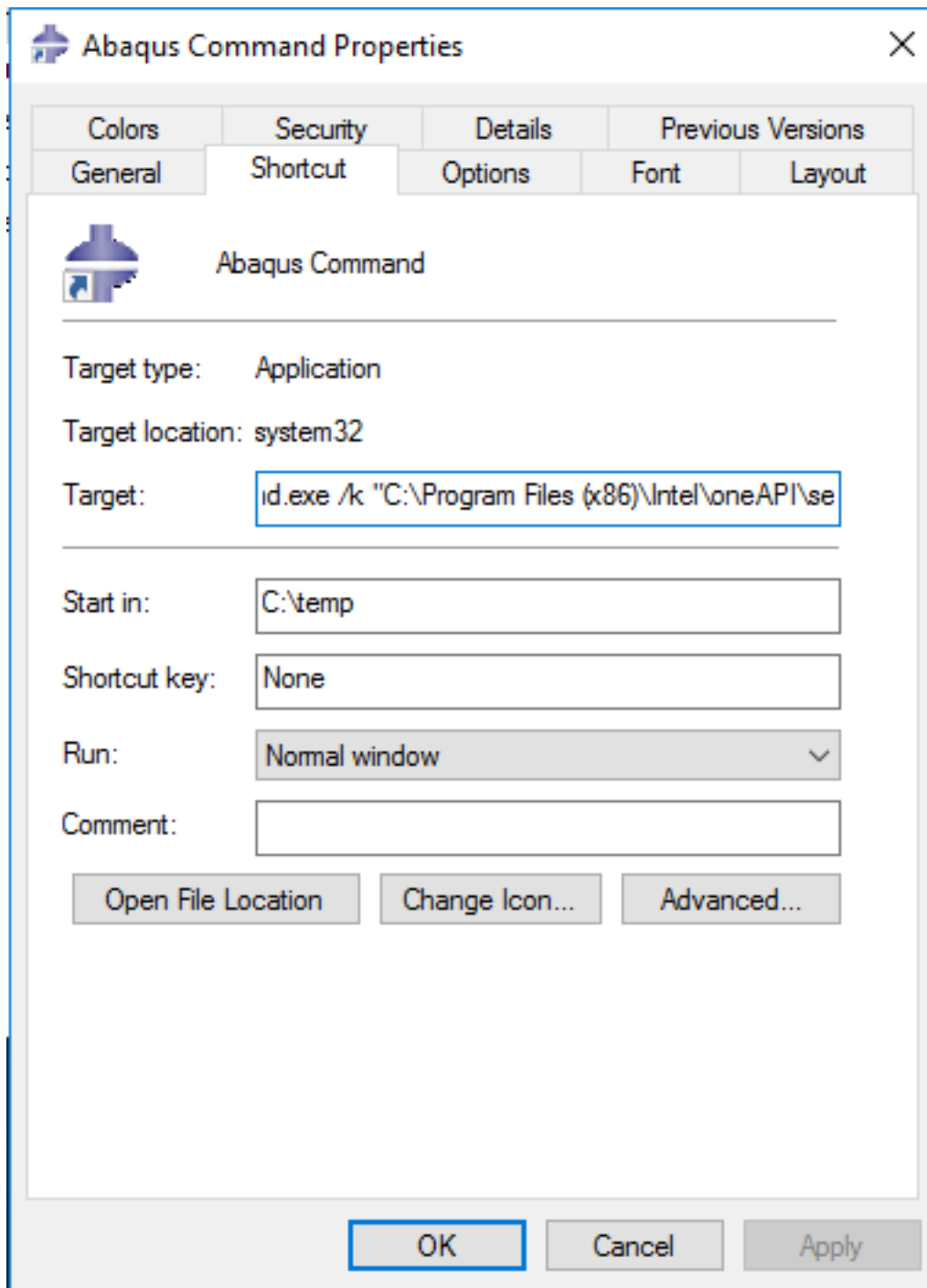
Step 3: Append Intel setvars.bat after /k in target field.



Step 4: Modify ABAQUS CAE shortcut target.



Step 5: Modify ABAQUS Verification shortcut target.



ABAQUS CAE Shortcut Command

"C:\Program Files (x86)\Intel\oneAPI\setvars.bat" && C:\SIMULIA\EstProducts\2021\win_b64\resour

ABAQUS Verification Shortcut Command

"C:\Program Files (x86)\Intel\oneAPI\setvars.bat" && C:\SIMULIA\EstProducts\2021\win_b64\resour

Abaqus CAE Properties

×

Colors

Security

Details

Previous Versions

General

Shortcut

Options

Font

Layout



Abaqus CAE

Target type: Windows Batch File

Target location: oneAPI

Target:

Start in:

Shortcut key:

Run:

Comment:

Open File Location

Change Icon...

Advanced...

OK

Cancel

Apply

Abaqus CAE Properties



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Abaqus CAE

Target type: Windows Batch File

Target location: oneAPI

Target: `call oneAPI\setvars.bat intel64 vs2019 && C:\SIMI`

Start in: `C:\temp`

Shortcut key: `None`

Run: `Normal window`

Comment: `launch Abaqus CAE`

Open File Location

Change Icon...

Advanced...

OK

Cancel

Apply

Abaqus Verification Properties

Colors

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General

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Font

Layout

Abaqus Verification

Target type: Windows Batch File

Target location: oneAPI

Target: `al\oneAPI\setvars.bat" intel64 vs2019 && C:\SIMI`

Start in: `C:\temp`

Shortcut key: `None`

Run: `Normal window`

Comment: `Abaqus verification`

Open File Location

Change Icon...

Advanced...

OK

Cancel

Apply

 Abaqus Verification Properties

Colors

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Previous Versions

General

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 Abaqus Verification

Target type: Windows Batch File

Target location: oneAPI

Target: `al\oneAPI\setvars.bat" intel64 vs2019 && C:\SIMI`

Start in: `C:\temp`

Shortcut key: `None`

Run: `Normal window`

Comment: `Abaqus verification`

Open File Location

Change Icon...

Advanced...

OK

Cancel

Apply

```
Intel(r) oneAPI Tools
:: initializing oneAPI environment...
:: initializing Visual Studio command-line environment...
Visual Studio version 16.8.4 environment configured.
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: ipp -- latest
: ippcp -- latest
: itac -- latest
: mkl -- latest
: mpi -- latest
: tbb -- latest
: vpl -- latest
: vtune -- latest
:: oneAPI environment initialized ::

C:\temp>
```

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